University of BRISTOL

COMS20018, Introduction to AI:
AI Ethics I:
Bias, Explainability & Privacy

Nan Lu

School of Computer Science
University of Bristol

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Agenda

1. **AI ethics overview**
   why, and why now
2. **Classifiers & errors**
   recap L03 + new metrics
3. **Bias**
   where it enters; big data paradox
4. **Explainability**
   LIME and SHAP
5. **Privacy**
   leakage, membership, DP
6. **Unlearning**
   making a model forget

Today's question

When an AI system makes a decision about you, how do we check which mistakes it makes, why it decided, and what it reveals about you?

A question to begin with

Which of these decisions about you could be made by an AI system?



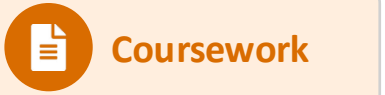
Admissions

"Will this applicant complete the degree?"



Placements

"Should we interview this CV?"



Coursework

"Was this essay written by AI?"



Online exams

"Is this student cheating?"

Each one is a **binary classifier**. Each of its mistakes lands on a real student.

What would you want to know before you trusted it?

In the EU AI Act, AI used for admissions, assessing students, monitoring tests, and screening job applicants is classed as high-risk (Annex III).

Source: EU AI Act, Annex III

AI ethics: the big picture

Common principles for responsible AI (UK AI regulation white paper, 2023, plus data protection)

Safety, security & robustness
works reliably, resists attacks — Week 11

Transparency & explainability
people can understand decisions — Today

Fairness
no unjustified differences between groups — Today + lect12

Privacy & data protection
UK GDPR: lawful use, right to erasure — Today

Accountability & governance
someone is responsible

Contestability & redress
you can challenge a decision

Why now?

- AI decisions are fast, cheap and repeated at scale: one flawed rule affects thousands.
- Models learn from historical data, including its biases.
- Complex models are hard to question.
- Models can memorise and leak their training data.

Principles are easy to agree on. This week: turning three of them into things we can compute, check and train with.

Source: UK AI regulation white paper (2023)

Recap (L03): logistic regression for classification

Running example: the placement screener

A company receives CVs from Bristol CS students for year-in-industry placements. A classifier decides who is shortlisted for a technical interview.

Features
$\phi(\mathbf{x})$
grades, projects, …

→

Probability
$y(\mathbf{x}, \mathbf{w}) = \sigma(\mathbf{w}^T \phi(\mathbf{x}))$

→

Threshold θ
$\hat{Y} = 1$ if $y(\mathbf{x}, \mathbf{w}) > \theta$
default θ = 0.5

→

Decision
Ŷ = 1 shortlist
Ŷ = 0 reject

Sigmoid

$$\sigma(a) = \frac{1}{1 + e^{-a}}$$

Training: minimise the cross-entropy loss

$$L(\mathbf{w}) = -\sum_{n=1}^{N} [t_n \ln y_n + (1 - t_n) \ln(1 - y_n)]$$

Notations

- **Y** = observed outcome
  (L03's label t; Y = 1: would pass the interview)
- $\hat{\mathbf{Y}}$ = the classifier's prediction
- $\mathbf{x}$: an applicant's input
- $\phi(\mathbf{x})$: its feature vector
- $\mathbf{w}$: learned weights
- θ: decision threshold
- $t_n \in \{0,1\}$: label of training example n
- $y_n = y(\mathbf{x}_n, \mathbf{w})$: its predicted probability;
- N: number of training examples.

Recap (L03): the confusion matrix

Positive ($\hat{Y} = 1$) = shortlisted. Rows: what is true (Y). Columns: what the model predicted ($\hat{Y}$).

	$\hat{Y} = 1$ (shortlisted)	$\hat{Y} = 0$ (rejected)
Y = 1 would pass	**True positive (TP)** shortlisted, and would pass	**False negative (FN)** rejected, but would have passed: missed talent
Y = 0 would not pass	**False positive (FP)** shortlisted, but would not pass: a wasted interview	**True negative (TN)** rejected, and would not pass

Reading the names

True / False: was the prediction correct?

Positive / Negative: what did the model predict, 1 or 0?

- Every applicant lands in exactly one cell: $\mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN} = N$ (the number of applicants).
- The two kinds of error (orange) harm different people.

Metrics from the confusion matrix

Metric	Formula	Question it answers for the screener	
Accuracy	$\frac{\mathrm{TP}+\mathrm{TN}}{N}$	How often is the screener right?	L03
Precision, also called positive predictive value (PPV)	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}$	Of those shortlisted, what fraction would pass?	L03
Recall = true positive rate (TPR)	$\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}$	Of students who would pass, what fraction are shortlisted?	L03
False negative rate (FNR)	$\frac{\mathrm{FN}}{\mathrm{TP}+\mathrm{FN}} = 1 - \mathrm{TPR}$	Of students who would pass, what fraction are rejected?	**new**
False positive rate (FPR)	$\frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}$	Of students who would not pass, what fraction are shortlisted?	**new**
Selection rate	$\frac{\mathrm{TP}+\mathrm{FP}}{N}$	What fraction of all applicants are shortlisted?	**new**

Notation: TP, FN, FP, TN: the four cell counts of the confusion matrix; N = TP + FN + FP + TN, the number of applicants. PPV = "positive predictive value": how far you can trust a positive prediction (the name comes from medical testing).

What each rate conditions on

TPR uses only the **Y = 1 row**; FPR only the **Y = 0 row**; precision only the **Ŷ = 1 column**: $\mathrm{TPR} = P(\hat{Y} = 1 \mid Y = 1)$, $\mathrm{FPR} = P(\hat{Y} = 1 \mid Y = 0)$.

Try it: scoring the screener

Test set: 200 applicants the screener has never seen

Placement screener, all applicants		
	$\hat{Y} = 1$	$\hat{Y} = 0$
Y = 1	60	20
Y = 0	18	102

Try it (3 min)

1. How many applicants would pass (Y = 1)? How many were shortlisted ($\hat{Y} = 1$)?

2. Calculate the accuracy, TPR, FPR, precision and selection rate (three decimal places).

3. A lazy screener rejects everyone. What are its accuracy and its TPR?

Try it: scoring the screener (answer)

1. Would pass: $\mathrm{TP} + \mathrm{FN} = 60 + 20 = 80$. Shortlisted: $\mathrm{TP} + \mathrm{FP} = 60 + 18 = 78$.

2. Formula, then numbers (TP = 60, FN = 20, FP = 18, TN = 102, N = 200):

$$\text{Accuracy} = \frac{\mathrm{TP} + \mathrm{TN}}{N} = \frac{60 + 102}{200} = 0.810$$

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = \frac{60}{60 + 20} = 0.750$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = \frac{18}{18 + 102} = 0.150$$

$$\text{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} = \frac{60}{60 + 18} \approx 0.769$$

$$\text{Selection rate} = \frac{\mathrm{TP} + \mathrm{FP}}{N} = \frac{60 + 18}{200} = 0.390$$

3. Reject everyone: TP = FP = 0, so $\text{Accuracy} = \frac{0+120}{200} = 0.600$ but $\mathrm{TPR} = \frac{0}{80} = 0$: it never finds anyone.

Take-away

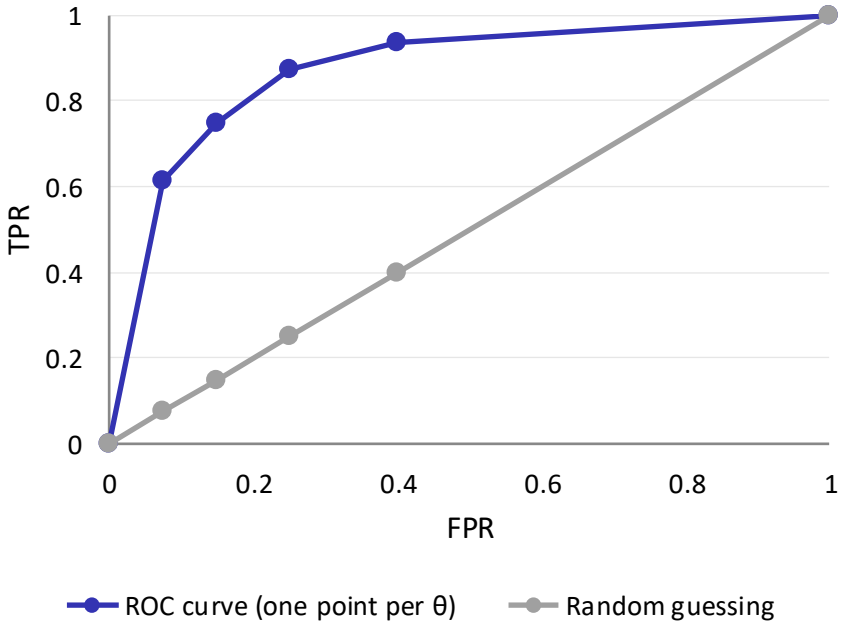
- Our screener misses 1 in 4 students who would pass (FNR = 0.25) and shortlists 15% of those who would not.
- Always read accuracy together with the confusion matrix and a simple baseline.

Moving the threshold θ: the ROC curve

ROC = Receiver Operating Characteristic (a name from radar signal detection)

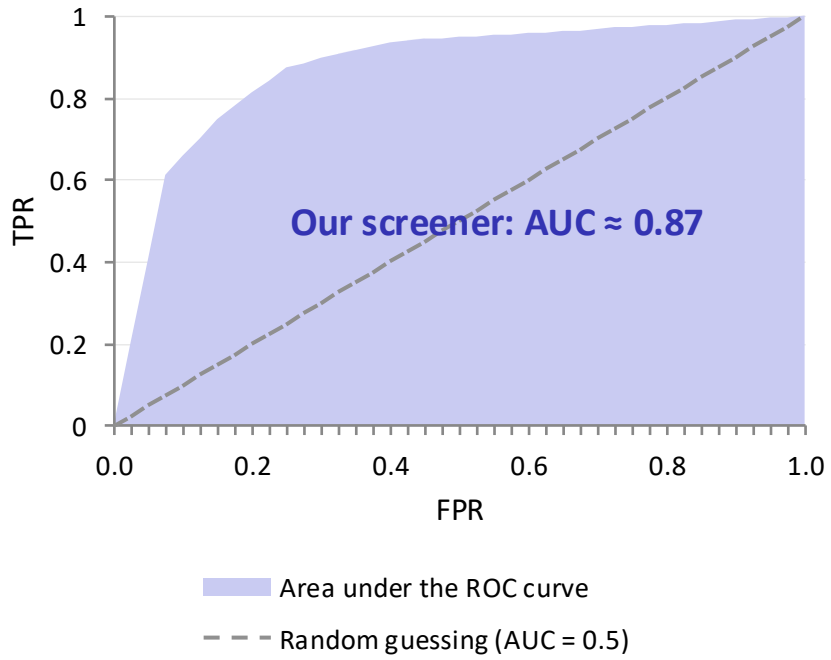
θ	TP	FP	TPR	FPR
0.6	49	9	0.613	0.075
0.5	60	18	**0.750**	**0.150**
0.4	70	30	0.875	0.250
0.3	75	48	0.938	0.400

Each threshold θ gives one confusion matrix, so one (FPR, TPR) point.



- **ROC curve**: plot TPR against FPR for every threshold θ, from θ = 1 (shortlist nobody: point (0, 0)) to θ = 0 (shortlist everybody: point (1, 1)).
- Lower θ: more students shortlisted, so TPR rises but FPR rises too.
- Top-left corner (FPR = 0, TPR = 1) is perfect; the diagonal is random guessing.

AUC: the Area Under the ROC Curve



What it measures

- One number for the whole ROC curve: 0.5 = random guessing, 1 = perfect.
- Threshold-free: compares models before θ is chosen.

Interpretation

$$\text{AUC} = P\big(s(\mathbf{x}^+) > s(\mathbf{x}^-)\big)$$

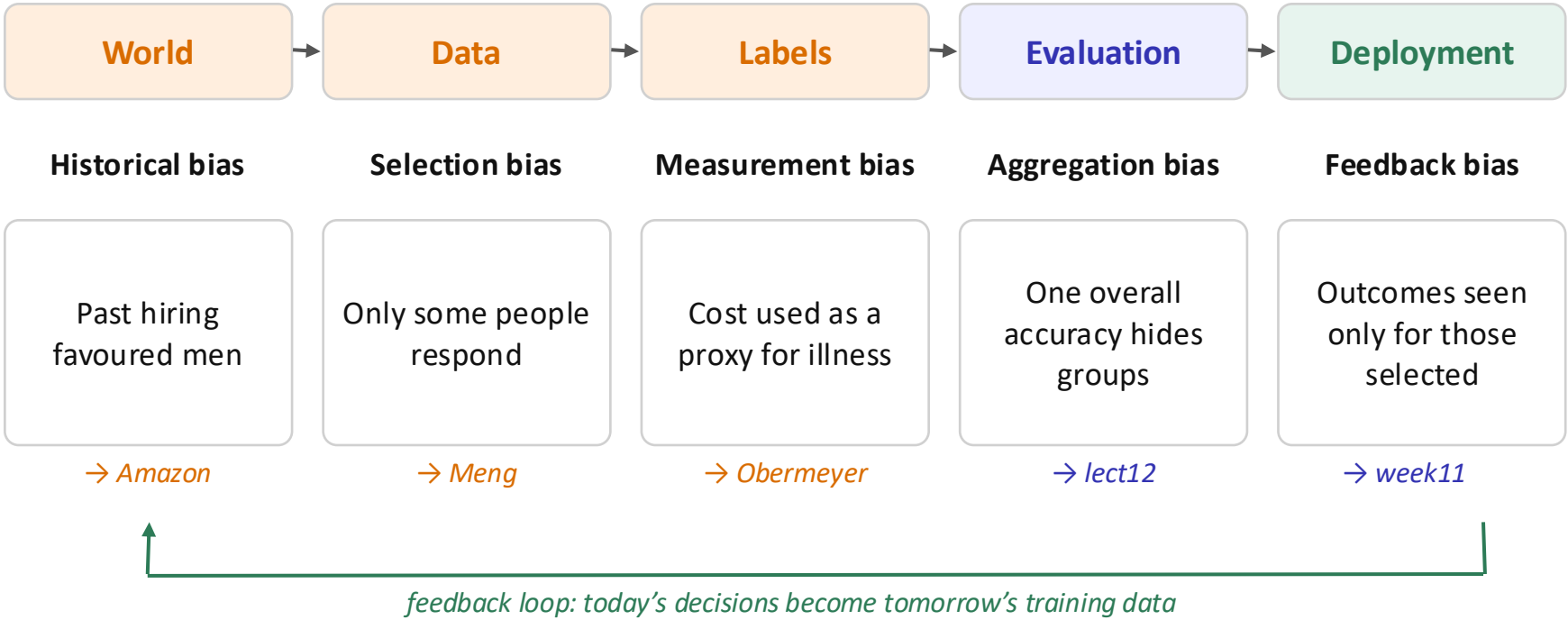
the probability that a randomly chosen positive gets a higher score than a randomly chosen negative.

Notations

- $s(\mathbf{x}) = y(\mathbf{x}, \mathbf{w})$: the model's score
- $\mathbf{x}^+$: a random applicant with Y = 1
- $\mathbf{x}^-$: a random applicant with Y = 0.

Bias: where does it enter?

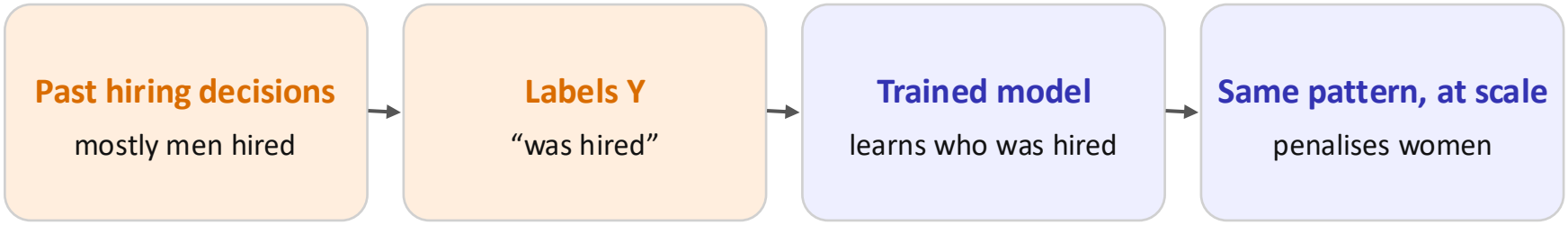
Bias can enter at every stage of the ML pipeline



A model can be biased even when it is trained correctly: it learns the world it is shown.

Source: Suresh & Guttag (2021)

Historical bias: Amazon's CV screener



- Trained on about 10 years of CVs submitted to Amazon, mostly from men.
- Learned to penalise the word "women's" (e.g. "women's chess club captain").
- Removing such words could not rule out other proxies; the project was dropped.

A model trained to imitate past decisions learns past bias, accurately.

Source: Reuters, Dastin (2018)

Measurement bias: a health algorithm that predicted cost

Obermeyer et al., Science (2019): a widely used US population-health algorithm

Goal
find patients who need extra care

→

Label actually used
predict future healthcare cost

→

The catch
at the same level of illness, Black patients historically incurred lower costs

→

Result
the model underestimated Black patients' health needs

17.7%
of patients auto-flagged for extra care were Black (cost label)

→

46.5%
if the bias were removed (label based on health)

Cost was less spent, not less need: unequal access to care.

The label Y is a choice. A convenient proxy can build bias into an accurate model.

Source: Obermeyer et al., Science (2019)

Selection bias at scale: the big data paradox

Meng (2018), Statistical Paradises and Paradoxes in Big Data: the 2016 US election

Race	Poll model	Hillary Clinton Democratic	Donald Trump Republican	Gary Johnson Libertarian	Jill Stein Green	Leading by (points)
Two-way	270toWin[4]	**47.2%**	43.6%	—		**3.6**
	BBC[5]	**48.0%**	44.0%			**4.0**
	HuffPost Pollster[6]	**47.3%**	42.0%			**5.3**
	New York Times[7]	**45.9%**	42.8%			**3.1**
	Real Clear Politics[8]	**46.8%**	43.6%			**3.2**
	TPM Polltracker[9]	**48.8%**	43.9%			**4.9**

Polls before the 2016 US presidential election: all predicted a Clinton lead.

> ***"The bigger the data, the surer we fool ourselves."***
>
> — Xiao-Li Meng

≈ 2,300,000

poll responses (≈ 1% of voters)

behaved like →

≈ 400

random responses

because

ρ ≈ −0.005

who responds vs how they vote

Sources: Meng (2018) · Wikipedia: 2016 US election polling

Why size does not save you: Meng's identity

$$\bar{G}_n - \bar{G}_N = \rho_{R,G} \times \sqrt{\frac{1-f}{f}} \times \sigma_G$$

Data quality

$\rho_{R,G}$

Data quantity

$f = n/N$

Problem difficulty

σ_G

Symbol	Meaning (2016 polls)
G_j	the value for person j, e.g. 1 if person j votes Trump, 0 otherwise
R_j	1 if person j is recorded (answers the poll), 0 otherwise
$\bar{G}_N$, $\bar{G}_n$	average of G over all N people (the truth) / over the n recorded people (the estimate)
$\rho_{R,G}$, σ_G	correlation between R and G / standard deviation of G across people

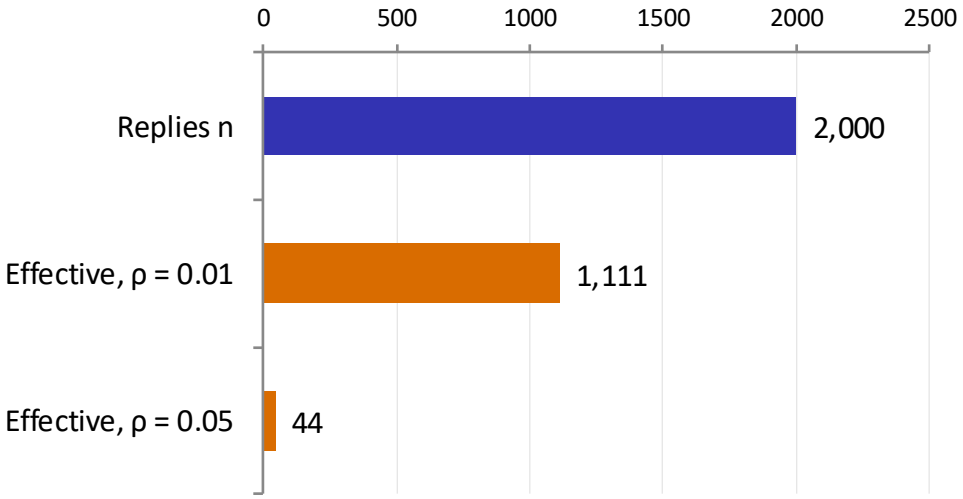
If $\rho \neq 0$, more data does not remove the error. Effective size: $n_{\text{eff}} \approx \frac{f}{1-f} \cdot \frac{1}{\rho^2}$

Source: Meng (2018)

Worked example: how much is a big survey worth?

The university emails a satisfaction survey to all N = 20,000 students; n = 2,000 reply. Satisfied students are slightly more likely to reply.

- Fraction recorded: $f = 2000/20000 = 0.1$, so $f/(1-f) \approx 0.111$.
- $\rho = 0.01$: $n_{\text{eff}} \approx 0.111/0.01^2 \approx 1111$ (2,000 replies ≈ 1,100 random ones)
- $\rho = 0.05$: $n_{\text{eff}} \approx 0.111/0.05^2 \approx 44$ (fewer than 50 random replies)
- Meng's election: $f = 0.01,\ \rho = -0.005 \Rightarrow n_{\text{eff}} \approx 0.0101/0.000025 \approx 404$



Take-away

- n_{eff} scales with $1/\rho^2$: 5× the correlation, 25× less information.
- A huge but self-selected training set can still give a biased model.

Source: Meng (2018)

Explainability: “why was I rejected?”

Evaluation (so far)

“How well does it perform?” Metrics on held-out data, for the model as a whole.

Explanation (now)

“Why did it decide this?” Which features moved the prediction, and in which direction.

	Global: the whole model	**Local: one decision**
Question	Which features matter across all applicants?	Why was this applicant rejected?
Tools	Logistic regression weights w_j (L03); average |SHAP|	LIME, SHAP: work for any black-box model

Why explanations matter for ethics

- **Contestability**: a rejected student can only challenge a decision they understand.
- **Finding bias**: reveals reliance on proxies, as in the Amazon case.
- **Trust**: a model can score well and still decide for reasons we would not accept.

Notation: w_j: the logistic-regression weight of feature j (how much it pushes the score up or down).

LIME: explain one decision with a simple model

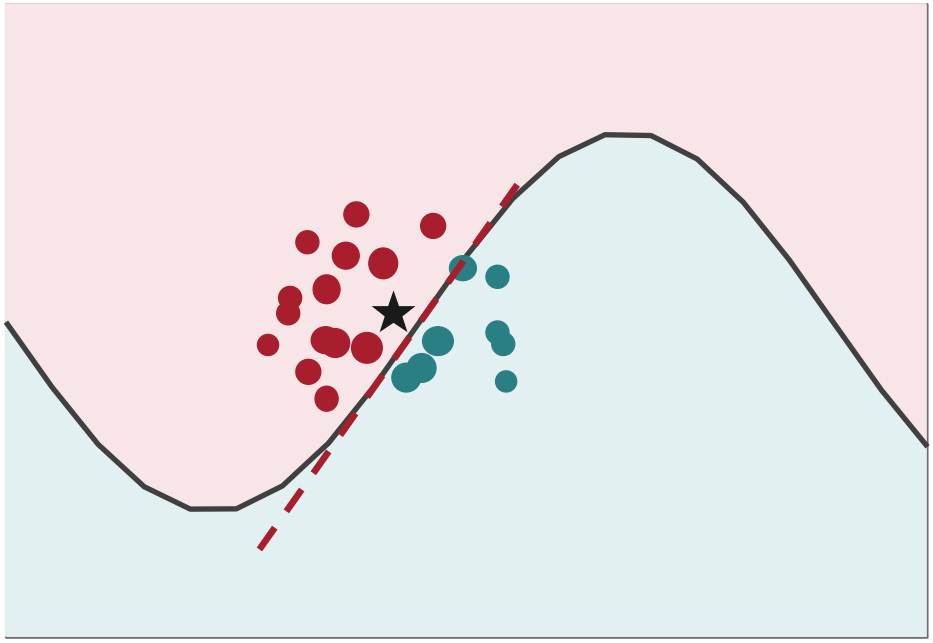
Local Interpretable Model-agnostic Explanations (LIME)

1 **Choose one case**
e.g. Sam's rejected CV (★)

2 **Create small variations**
change one or two features at a time (dots)

3 **Ask the black-box model**
get its prediction for each variation

4 **Fit a simple model nearby**
a straight line that copies the black box near ★ (dashed)



★ case explained · dots = variations · dashed = simple local model; solid = true boundary

Idea: the true decision boundary may be curved, but close to one case it looks almost straight. The weights of that local straight line are the explanation.

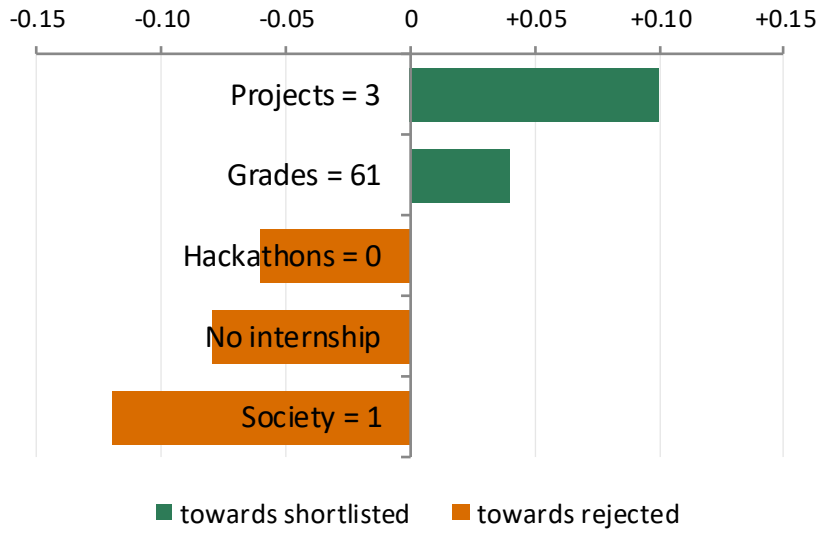
Sources: Ribeiro et al. (2016), LIME · Molnar, Interpretable ML

LIME for Sam, step by step

The screener gives Sam $y(\mathbf{x}, \mathbf{w}) = 0.38 < \theta = 0.5$, so Sam is rejected.

CV	What changed	P(shortlist)
Sam	nothing (original)	**0.38**
Variation 1	society removed	0.50
Variation 2	4 projects instead of 3	0.47
Variation 3	has an internship	0.46
Variation 4	grade average 55	0.35
…	500 variations in total	…

Steps 2–3: vary Sam's CV and ask the screener



Step 4: weights of the fitted straight line

What it reveals: the biggest push towards rejection is membership of a "Women in Tech" society, a proxy for gender. The same problem as Amazon's tool, found by explaining a single decision.

All values are illustrative. The weights explain this one decision, not the model everywhere.

Source: Ribeiro et al. (2016), LIME

SHAP: share the prediction fairly among features

SHapley Additive exPlanations (SHAP), based on Shapley values from game theory

Game theory

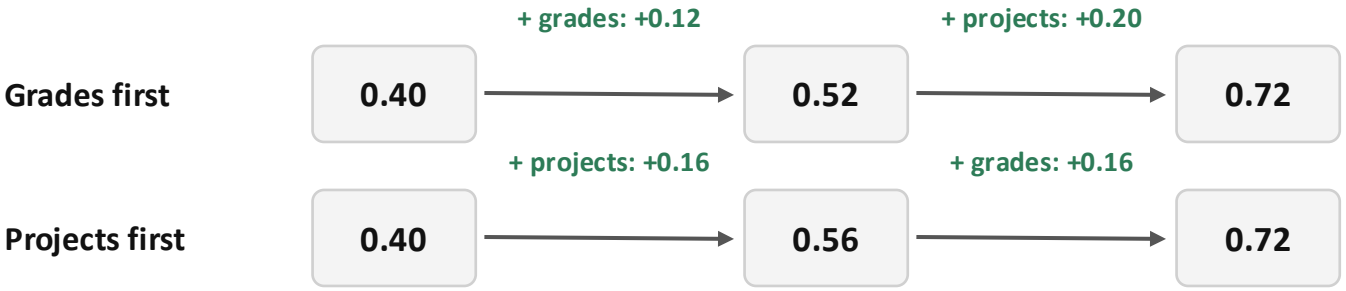
Players team up and win a prize. How do we split it fairly when some players add more than others?

SHAP

Features are the players; the prize is how far this prediction is from the average prediction.

A feature's credit = its average extra contribution over every order in which features could be added

Example with two features: start from the average applicant (0.40), add features one at a time.



Average credit

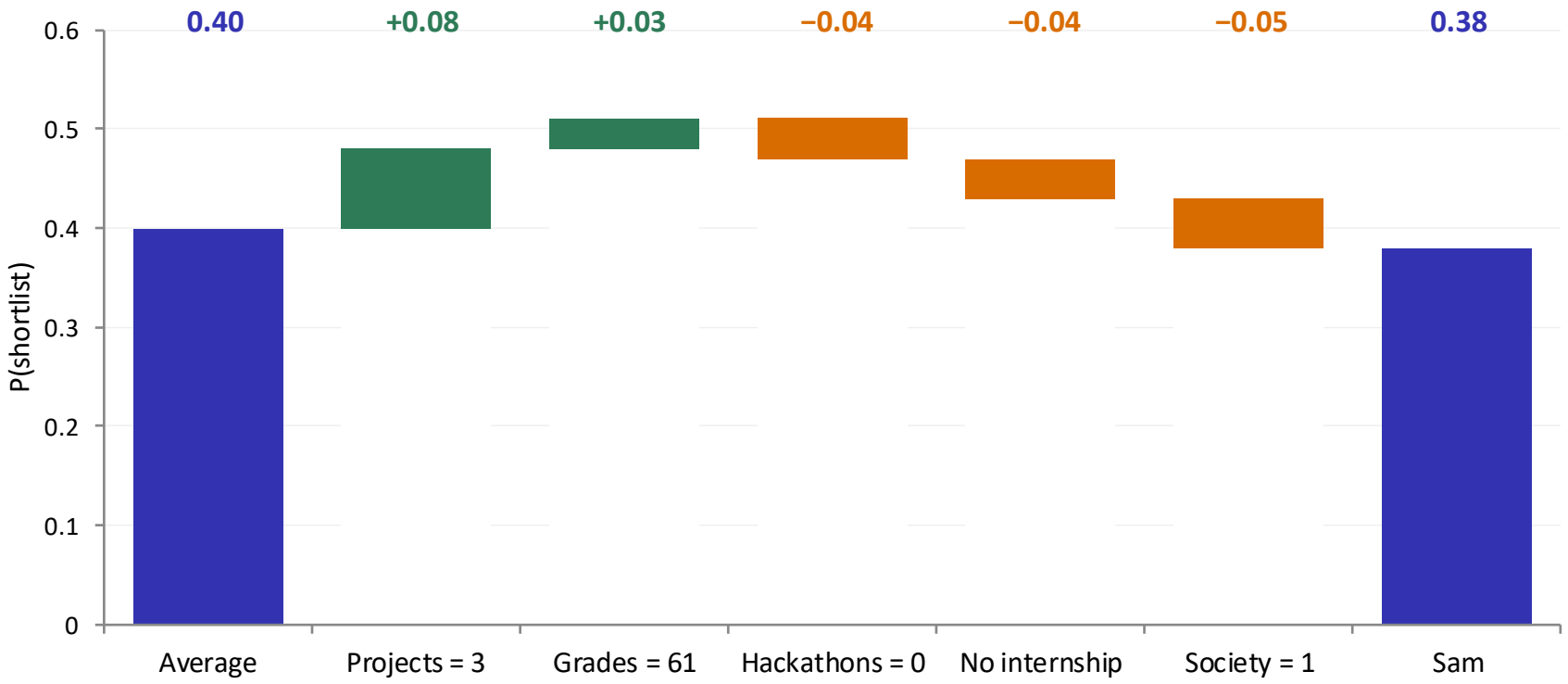
grades: (0.12 + 0.16)/2 = 0.14

projects: (0.20 + 0.16)/2 = 0.18

The credits add up: $0.40 + 0.14 + 0.18 = 0.72$ (average + credits = prediction).

Sources: Lundberg & Lee (2017), SHAP · Molnar, Interpretable ML

SHAP for Sam: from the average to Sam's prediction



- **Start** at the average applicant's probability, 0.40.
- **Each bar** is one feature's credit for Sam: green pushes towards shortlisting, orange towards rejection.
- **End** at Sam's probability, 0.38. The society membership is again a strong push towards rejection.

$$0.40 + 0.08 + 0.03 - 0.04 - 0.04 - 0.05 = 0.38$$

Source: Lundberg & Lee (2017), SHAP

LIME vs SHAP, and what explanations do not show

	LIME	SHAP
Core idea	fit a simple model near the case (local: single instance)	average contribution over orders (both local and global: single/overall behavior)
Adds up to the prediction?	not guaranteed	yes, by construction
Stability	depends on the variations; can vary	consistent, theoretically grounded
Cost	cheap	expensive exactly; fast approximations

Limits

- Describe the model, not causes in the world
- Do not show that a decision is correct or fair
- Can be manipulated to hide bias (Slack et al., 2020)



Explanations can leak

Detailed explanations reveal information about the training data, and can make membership attacks easier (Shokri et al., 2021).

Transparency and privacy can pull against each other.

Next: what a model reveals about the people it was trained on.

Sources: Slack et al. (2020) · Shokri et al. (2021)

Privacy: “anonymous” data often isn’t

87%

of the US population is likely unique on {ZIP code, gender, date of birth}



Linkage re-identification

- Massachusetts released “anonymised” hospital records: names removed, ZIP + gender + birth date kept.
- Joined with a public voter list, they revealed the Governor’s own medical record.

Closer to home: an anonymous unit survey

Programme	Year	Gender	Home country	Comment
CS with AI	2	F	Norway	“The lectures are too hard…”

- No name, but in a cohort of ~40 this row may match exactly one person.
- **Quasi-identifiers**: attributes that are harmless alone but identifying together.

Removing names is not anonymisation.

Source: Sweeney (2000)

A trained model can leak too: three routes back to you

1 Linkage

Anonymous records + public data → identity

Example: *Survey row + LinkedIn → you*

2 Membership inference

Query the model → was this person IN the training data?

Example: *Was my record in the "at-risk student" model?*

3 Extraction / inversion

Query the model → recover memorised records or attributes

Example: *An LLM reproduces an email address from its training text*

Why models leak

- Models fit their training data better than new data (overfitting, L03): lower loss, higher confidence on members.
- **Deleting a row from the database does not delete it from the model.**

Sources: Shokri et al. (2017) · Carlini et al. (2021)

Formulating privacy: the membership game

To measure privacy, define what an attacker is trying to do

1 Train

A model f is trained on dataset D.

2 Challenge

Flip a coin: pick a record z from D (IN) or a fresh record from the same population (OUT).

3 Guess

The attacker sees f(z), e.g. the model's confidence, and guesses IN or OUT.

The attacker is a binary classifier (positive = IN)

- **Simple attack:** guess IN if the model's loss on z is low (members have lower loss).
- **Membership advantage:** $\mathrm{Adv} = \mathrm{TPR} - \mathrm{FPR}$
- **Example:** says IN for 70% of members and 30% of non-members: $\mathrm{Adv} = 0.70 - 0.30 = 0.40$.
- Adv = 0: no better than a coin flip, so the model reveals nothing about membership.

Notation: TPR = fraction of true members (IN) the attack calls IN; FPR = fraction of non-members (OUT) it calls IN; f(z) = the model's output on record z.

Sources: Yeom et al. (2018) · Shokri et al. (2017)

Differential privacy: a guarantee, not a hope

Idea: the model's output should barely change whether your record is included or not

D: all students, including you

D': the same, without you

(neighbouring datasets)

→

A randomised algorithm M is **ε-differentially private** if for all neighbouring D, D' and all sets of outputs S:

$$P[M(D) \in S] \leq e^{\varepsilon} P[M(D') \in S]$$

- **Small ε = strong privacy**.
- ε = 0: the output does not depend on you at all.
- Membership inference is bounded: $\mathrm{Adv} \leq e^{\varepsilon} - 1$.

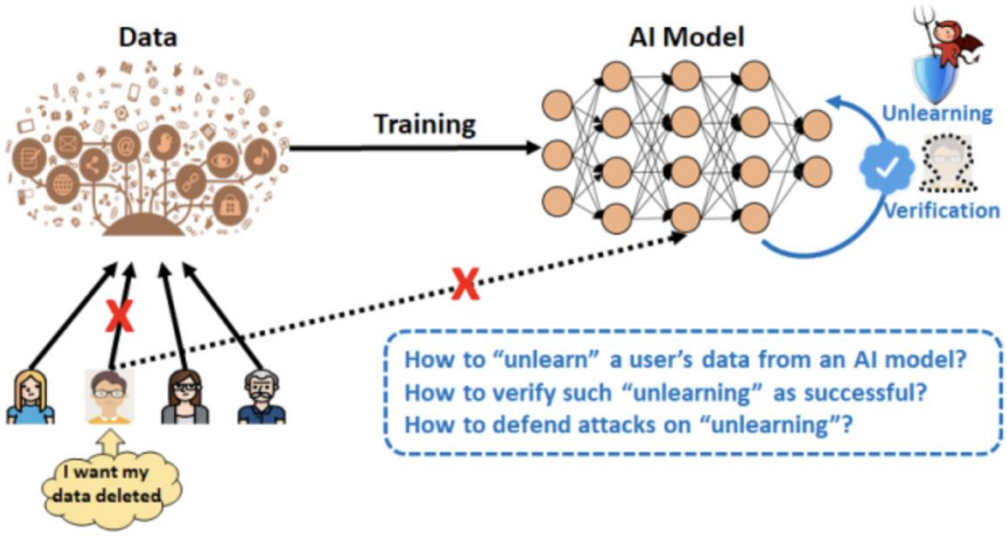
Notations

- M: the algorithm (e.g. training a model), which uses randomness
- M(D): its output on data D
- S: any set of possible outputs
- ε ≥ 0: the privacy level (smaller = more private).

Sources: Dwork & Roth (2014) · Yeom et al. (2018) · Abadi et al. (2016), DP-SGD

Machine unlearning: making a model forget

Why: UK GDPR Article 17, the right to erasure; withdrawn consent; wrong or poisoned records.



A good unlearning method gives

Forgetting
deleted data look unseen

Utility
accuracy is kept

Efficiency
much cheaper than retraining

	Formulation
Data	Training data D, split into D_f (to forget, e.g. students who asked to be deleted) and $D_r = D \backslash D_f$ (to keep).
Oracle	The model retrained from scratch on D_r only: the gold standard, but usually far too expensive.
Goal	Starting from the trained model, cheaply get a model that behaves like the oracle (same predictions, same leakage about D_f).

Sources: ICO: right to erasure · Cao & Yang (2015) · Nguyen et al. (2022) survey

Mainstream unlearning methods: the ideas

Method	Idea in one line	Guarantee / cost
Retrain from scratch	Train again on D_r only (the oracle).	exact · slowest
Fine-tune	Keep training on D_r only, so the model gradually forgets D_f.	no guarantee · cheap
Gradient ascent	Increase the loss on D_f (undo its learning), then briefly fine-tune on D_r to repair accuracy.	no guarantee · cheap; common for LLMs
Random relabelling	Train briefly on D_f with wrong, random labels so its patterns are overwritten.	no guarantee · cheap
Influence update	Estimate how much D_f moved the weights and subtract it in one step (+ noise).	certified for simple models such as logistic regression

Did it really forget? Play the membership game on the forgotten records

After unlearning, the attack should say "IN" for forgotten records no more often than for never-seen records:

$\mathrm{TPR}_{D_f} \approx \mathrm{FPR}$. **Unchanged test accuracy is not evidence of forgetting.**

Notation: TPR_{D_f}: fraction of forgotten records the attack still calls IN; FPR: fraction of never-seen records it calls IN.

Sources: Nguyen et al. (2022) survey · Graves et al. (2021) · Guo et al. (2020)

Summary

Five things to remember

1. The confusion matrix gives accuracy, TPR, FPR, precision and selection rate; the ROC curve and AUC summarise all thresholds.

2. Bias enters through the world (Amazon), data (Meng), labels (Obermeyer), evaluation and deployment.

3. Big self-selected data can be worth little: “The bigger the data, the surer we fool ourselves.”

4. LIME and SHAP explain single decisions and can expose proxies, but describe the model, not causes.

5. Privacy can be measured (membership advantage), guaranteed (differential privacy) and restored (unlearning).

Reading

Core: Statistical Paradises and Paradoxes in Big Data (I)
X.-L. Meng (2018), Annals of Applied Statistics 12(2). Focus: Sections 1–4 and the election example.

Dissecting racial bias in an algorithm used to manage the health of populations
Obermeyer et al. (2019), Science 366.

Interpretable Machine Learning: LIME and SHAP chapters
C. Molnar. Free online book.

Fairness and Machine Learning, Ch. 1 (Introduction)
Barocas, Hardt & Narayanan. Free online; read before tomorrow.

The Algorithmic Foundations of Differential Privacy, Ch. 2–3
Dwork & Roth (2014). Includes randomised response.

A Survey of Machine Unlearning
Nguyen et al. (2022).